



Pacia, Jasmin Clara S.

BSIT 4A_WAMD

WEEK 2

Part 1 – COMPANY PROFILE

1.1 Company Name

ABC Startup Solutions

1.2 Nature of Business

ABC Startup Solutions is a newly founded software development firm that develops digital solutions for companies and institutions. Its primary tasks include maintaining software systems, creating web and mobile applications, and offering clients technology-based solutions that can enhance their day-to-day operations.

Information Technology, Human Resources, Finance, and Sales are the four departments that make up the company's operations since it has just 20 employees. In order to support the company's software development activities and general business operations each department plays a distinct role.

1.3 Company Vision

To establish our company as a reputable software development firm that consistently provides practical digital solutions while constantly adapting to the changing needs of businesses and the technology industry.

1.4 Company Mission

ABC Startup Solutions is dedicated to creating useful and reliable software solutions that meet the needs of its clients. The company hopes to accomplish this by combining technical expertise, teamwork, responsible use of technology, and continuous improvement, all while providing employees with an environment in which they can grow and contribute to the organization's success.

1.5 Office Location

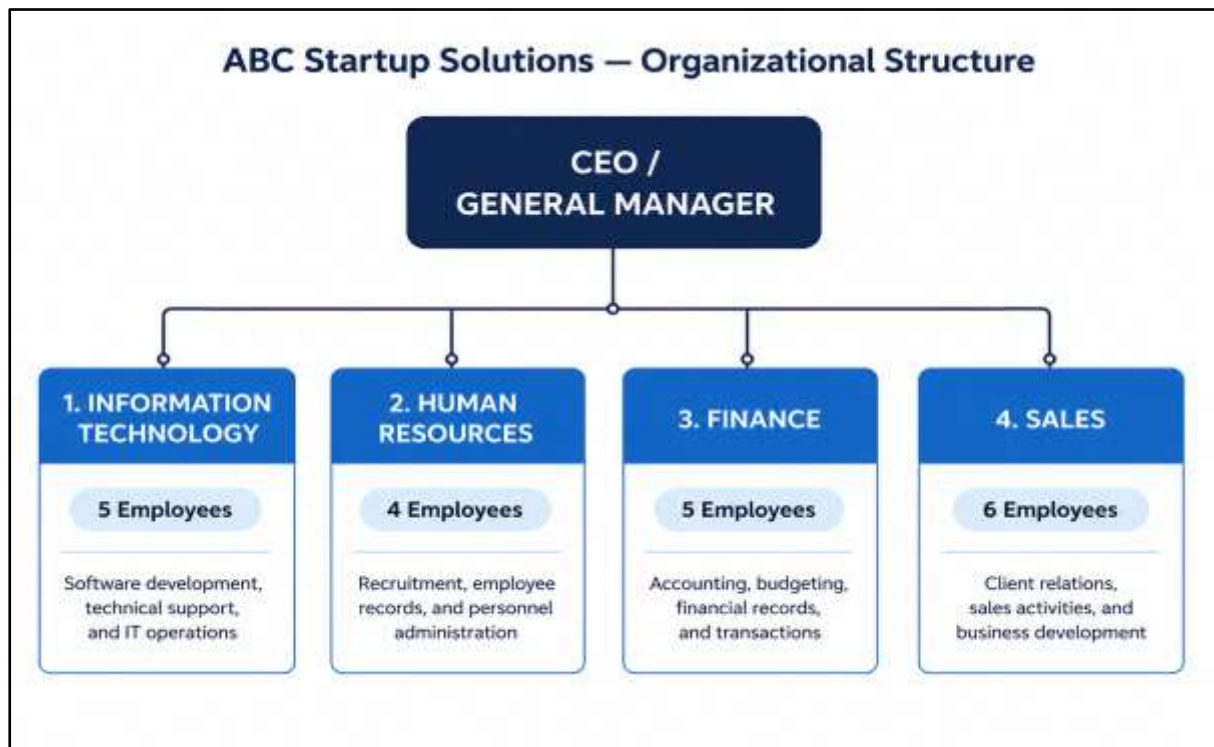
ABC Startup Solutions

2nd Floor, ABC Business Center

National Highway, Los Baños, Laguna, Philippines

1.6 Organizational Structure

ABC Startup Solutions employs a functional organizational structure in which employees are grouped based on their responsibilities. The CEO, also known as the General Manager, is in charge of overseeing the company and coordinating the major departments. The Information Technology, Human Resources, Finance, and Sales departments are each responsible for a specific function and collaborate to support the company's operations.



1.7 Employee Distribution

ABS Startup Solutions company has a total of 20 employees divided into four departments. The Information Technology department has five employees who work on software development and IT operations. Human Resources has four employees who handle employee-related and administrative tasks. Finance has five people to manage financial matters, while Sales has six people to handle client relations and business development.

Department	Number of Employees	Main Function
Information Technology	5	- Software development - Technical Support - IT Operations - System Administration and Maintenance
Human Resources	4	- Recruitment - Training and Workshops - Employee Records - Personnel Administration
Finance	5	- Accounting - Budgeting - Financial Records - Transactions
Sales	6	- Client Relations - Sales Activities - Business Development - Marketing
Total	20	Overall Company Workforce



PART 2

2.1 Enterprise Hardware Inventory

Asset ID	Hardware	Quantity	Department	Purpose
HW-001	Desktop Computers	15	IT, HR, Finance & Sales	Used as primary workstations for employees who perform their regular duties within the office.
HW-002	Laptops	5	IT	Used by IT personnel for software development, system administration, troubleshooting, and technical work that may require mobility.
HW-003	Server	1	IT	Provides centralized computing services and supports the company's internal IT environment.
HW-004	Router	1	IT	Connects the company's internal network to the internet and manages network traffic.
HW-005	Switch	1	IT	Provides wired network connectivity between computers, the server, printer, and other network devices.
HW-006	Printer	2	HR & Finance	Provides shared printing for employee documents, administrative records, reports, and financial documents.
HW-007	UPS	4		Provides power protection for important infrastructure equipment during power interruptions or fluctuations.
HW-008	Wireless Access Point	2	Entire Office	Provides wireless network connectivity throughout the office for authorized wireless devices.
HW-009	NAS Storage	1	IT	Provides centralized network storage for shared files and company data.
HW-010	External Backup Drive	2	IT	Provides additional backup storage for important company data and allows backup copies to be kept separately from the main systems.
HW-011	Monitors	20	IT, HR, Finance & Sales	Provides a dedicated display and tool for each employee's for doing their job on their assigned workstation.



Part 3

3.1 Enterprise Software Inventory

ABC Startup Solutions' software inventory lists the software required to support the company's operating systems, software development activities, productivity requirements, system administration, web access, security, remote support, and file management.

Software	Version	License	Purpose
Windows 11 Pro	25H2 (Build 26200.8875)	Commercial	<i>Operating system for the company's desktop computers and laptops</i>
Ubuntu Server	26.04 LTS	Open Source	<i>Operating system for the company's server</i>
Microsoft Office	Version 2606 (Build 20131.20154)	Commercial	<i>Productivity software for documents, spreadsheets, presentations, and other office tasks</i>
VS Code	1.130	Free	<i>Source code editor for software development</i>
Git	2.55.0	Open Source	<i>Version control for managing and tracking source-code changes</i>
GitHub Desktop	3.6.3	Free	<i>Graphical application for managing Git and GitHub repositories</i>
VirtualBox	7.2.14	Free / Open Source	<i>Graphical application for managing Git and GitHub repositories</i>
Google Chrome	150.0.7871.181/.182	Free	<i>Web browser for accessing websites and web-based services</i>
Microsoft Defender	Platform 4.18.26020.6	Included	<i>Security protection for Windows devices</i>
AnyDesk	9.7.12	Free / Commercial	<i>Remote access and technical support</i>
7-Zip	26.02	Open Source	<i>File compression and extraction</i>



Why each of this software is needed?

- **Windows 11 Pro** will serve as the primary operating system for the company's desktop computers and laptops. It provides the environment needed for employees to run their workplace applications and perform their daily tasks.
- **Ubuntu Server** will be installed on the company's server. The LTS release provides the server operating system required for the company's server environment.
- **Microsoft Office** will support regular office work such as preparing documents, creating spreadsheets, and making presentations.
- **Visual Studio Code** will be used by the IT department for writing and editing source code as part of the company's software development activities.
- **Git** will be used for version control so developers can track and manage changes made to source code.
- **GitHub Desktop** will provide a graphical interface for working with Git repositories and GitHub-based development projects.
- **VirtualBox** will allow IT personnel to create and run virtual machines for development and system administration activities.
- **Google Chrome** will provide employees with a web browser for accessing websites, web applications, and other online services.
- **Microsoft Defender** will provide security protection for the company's Windows devices against malware and other threats.
- **AnyDesk** will allow authorized IT personnel to provide remote technical assistance and troubleshoot supported computers.
- **7-Zip** will be used to compress and extract files and manage compressed archives.

Part 4

4.1 Enterprise Network Inventory

ABC Startup Solutions requires a basic network infrastructure to connect its employees, servers, and other network devices within the office. The following inventory identifies the network equipment needed to establish reliable wired and wireless connectivity and support the company's daily operations.

Equipment	Quantity	Purpose
ISP Modem	1	Provides the connection between the company's network and the Internet Service Provider.
Router	1	Routes network traffic between the company's internal network and external networks.
Firewall	1	Controls and filters network traffic to help protect the company's internal network from unauthorized access.
Switch	1	Connects the company's wired computers, server, printer, and other network devices.
Access Point	2	Provides wireless network connectivity throughout the office.

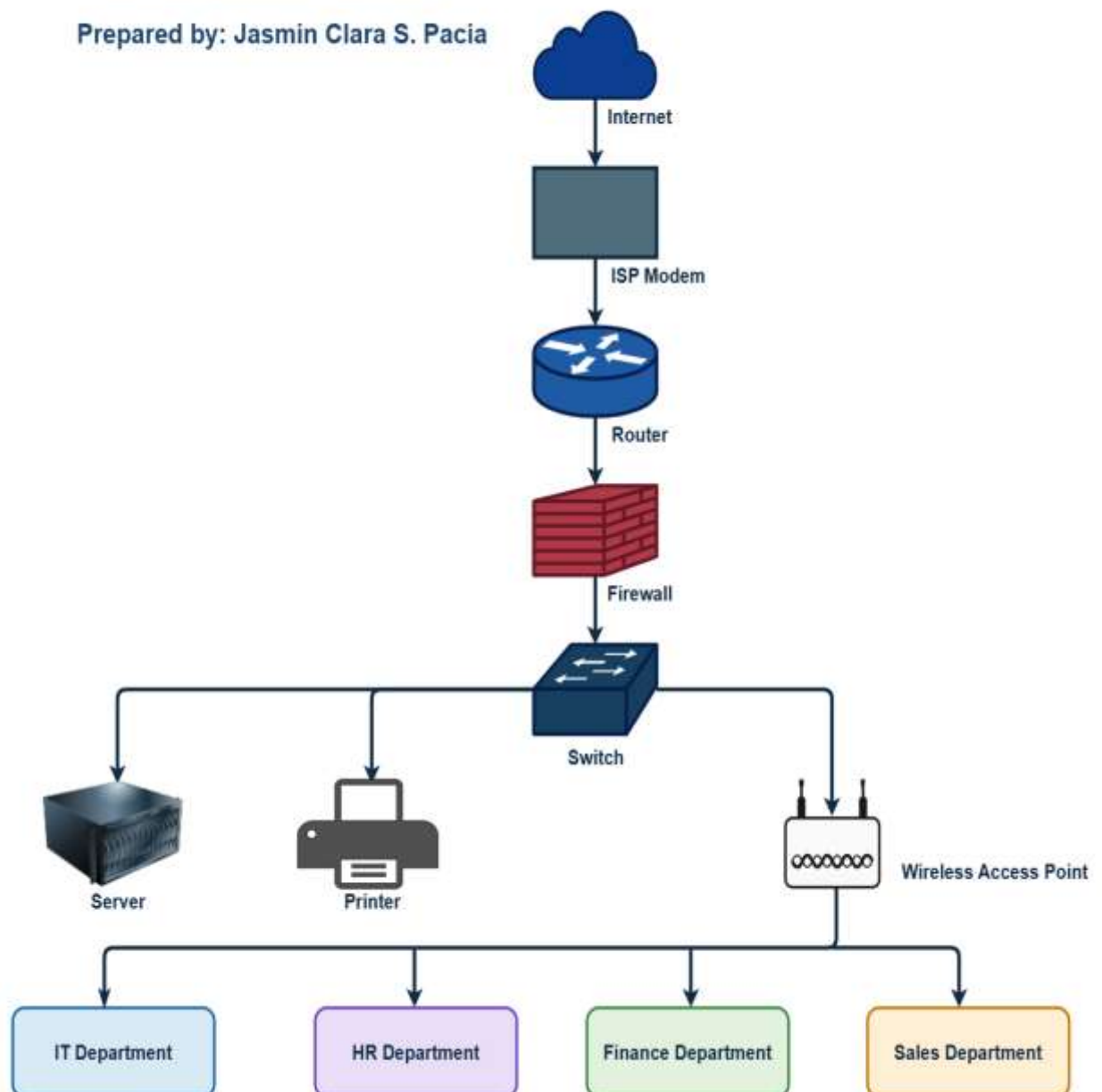


Patch Panel	1	Organizes and manages the Ethernet connections between office network cables and the network switch.
CAT6 Cables	25	Provides wired Ethernet connections between computers and network equipment.
RJ54 Connectors	50	Used to terminate CAT6 Ethernet cables and connect them to network ports.

Part 5

5.1 Enterprise Network Diagram

ABC Startup Solutions
Enterprise Network Topology
Prepared by: Jasmin Clara S. Pacia





Part 6

6.1 System Administration Roles

Helpdesk Technician

- **Responsibilities:** Handles user concerns, troubleshoots basic hardware and software issues, manages support requests, and escalates complex problems.
- **Skills:** Troubleshooting, communication, customer service, basic networking, and documentation.
- **Common Tools:** Windows, Microsoft 365, Command Prompt, PowerShell, and helpdesk systems.
- **Certifications:** CompTIA A+ and CompTIA Network+.

Network Administrator

- **Responsibilities:** Maintains the organization's network, including routers, switches, firewalls, and wireless devices.
- **Skills:** Networking, troubleshooting, routing and switching, network security, and monitoring.
- **Common Tools:** Cisco devices, Wireshark, SSH, Ping, and Traceroute.
- **Certifications:** CCNA and CompTIA Network+.

Linux System Administrator

- **Responsibilities:** Manages Linux servers, users, permissions, services, updates, and system performance.
- **Skills:** Linux administration, command-line operation, Bash, troubleshooting, and system security.
- **Common Tools:** Ubuntu Server, Bash, SSH, systemctl, and apt.
- **Certifications:** RHCSA and CompTIA Linux+.

Cloud Administrator

- **Responsibilities:** Manages cloud resources, user access, monitoring, security, and cloud services.
- **Skills:** Cloud computing, identity management, networking, security, and troubleshooting.
- **Common Tools:** AWS, Microsoft Azure, AWS Management Console, and Azure Portal.
- **Certifications:** AWS Certified Cloud Practitioner and Azure Administrator Associate

The four professionals work together to keep the organization's IT environment reliable and available to employees. The **Helpdesk Technician** handles initial user concerns and resolves common technical problems, while more complex network-related issues can be escalated to the **Network Administrator**. The **Linux System Administrator** manages



Linux servers and ensures that server-based services remain functional, while the **Cloud Administrator** manages the organization's cloud resources, access, and cloud-based services. By coordinating their responsibilities and sharing information about technical issues, these professionals help maintain an organized IT environment and reduce interruptions to the organization's daily operations.

Part 7

7.1 Infrastructure Recommendations

1. Internet Provider

- **Recommendation:** Use PLDT Enterprise Fiber Internet with a 500 Mbps business plan as the company's primary Internet connection.
- **Reason:** A 500 Mbps business connection provides sufficient bandwidth for cloud applications, file transfers, communication, and simultaneous Internet use by employees.

2. Server Specifications

- **Recommendation:** Use an HPE ProLiant ML30 Gen11 server with an Intel Xeon E-2436 processor, 32 GB ECC RAM, and SSD storage configured for redundancy.
- **Reason:** This provides sufficient processing capacity and memory for company services while ECC memory and redundant storage improve server reliability.

3. Backup Strategy

- **Recommendation:** Implement a 3-2-1 backup strategy using a Synology NAS, an external backup drive, and one cloud backup copy.
- **Reason:** Keeping multiple copies across different storage locations reduces the risk of losing important company data because of hardware failure, accidental deletion, or other incidents.

4. Security Recommendations

- **Recommendation:** Deploy a Fortinet FortiGate firewall, enable Multi-Factor Authentication (MFA) for company accounts, and apply role-based access control.
- **Reason:** These controls help protect the network from unauthorized access while ensuring employees only receive access appropriate to their responsibilities.

5. Antivirus

- **Recommendation:** Use Microsoft Defender for Business for company computers and supported devices.



- **Reason:** It provides centralized endpoint protection against malware and other security threats while integrating with the organization's Windows environment.

6. Password Policy

- **Recommendation:** Require passwords with a minimum of 12 characters, prohibit password sharing, and enable MFA for accounts that support it.
- **Reason:** Longer passwords and MFA provide stronger protection against unauthorized account access.

7. Expansion Plan

- **Recommendation:** Design the network using a managed Gigabit Ethernet switch with additional available ports and maintain expandable server and storage capacity.
- **Reason:** This allows ABC Startup Solutions to add employees, computers, printers, and other network devices without immediately replacing its core network infrastructure.

Part 8

8.1 Reflection

This project really opened my eyes to what system administration's all about. I found out that it is not about setting up computers and networks. I learned that planning the IT infrastructure is super important before you even buy any equipment or set up any systems. When I created the company profile I had to make lists of all the hardware and software. I had to map out the network. This helped me see how all the different parts of the infrastructure work together to support the organization. I also learned that when you are choosing hardware, software and network devices you have to think about the people who work at the company the departments how things are done and what the company needs in the future. The hardest part for me was making the enterprise network diagram. I had to make sure I included all the devices and that they were connected in a way that made sense. I also had to figure out how to use Draw.io to make a looking diagram with the right symbols and labels that people could read. This task taught me that even a simple network diagram needs to be planned because if the connections are wrong or the layout is confusing it can be hard to understand the infrastructure.

Planning is crucial before you start setting things up because it helps you avoid wasting money having problems with compatibility, security issues and making decisions, about the infrastructure. If you plan first a System Administrator can figure out what the organization needs choose the equipment and software and decide how all the different systems will work together. Planning also makes it easier to add to the infrastructure when the company gets bigger.

This project will help me become a better System Administrator because it allowed me to practice technical planning, documentation, inventory management, network design, and decision-making. It also improved my ability to communicate technical information in an organized and professional way. More importantly, I learned that a System Administrator must



Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna
COLLEGE OF COMPUTER STUDIES



think beyond individual devices and consider how the entire IT infrastructure works as one system. These skills will be useful in future projects where I will need to design, deploy, maintain, and troubleshoot real-world IT environments.